

QC01_all_correct

Every compared global-embedding system answers correctly. Development set, train_40k, seed 0. Condition: normal.

question-only head: a classifier over the frozen CLIP question embedding alone, with no image input; concatenation head: a classifier over the frozen CLIP image and question embeddings concatenated; fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings.



image 2384466

Question: Is the man to the right of a boy?

Gold answer: no

Question-only: no

✓ correct

Concatenation: no

✓ correct

Fusion: no

✓ correct

Illustrative only. An easy row for every system; illustrative of the task, not of any system's advantage.



image 2340934

Question: Are there either any shelves or computers?

Gold answer: yes

Question-only: yes

✓ correct

Concatenation: yes

✓ correct

Fusion: yes

✓ correct

Illustrative only. An easy row for every system; illustrative of the task, not of any system's advantage.

QC04_representation_improvement

The SigLIP fusion head correct where the CLIP fusion head is wrong. Development set, train_40k, seed 0. Condition: normal.

fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings; SigLIP fusion head: the same fusion head over frozen SigLIP ViT-B/16 embeddings instead of frozen CLIP ViT-B/32.



image 2318195

Question: Is the yellow food to the left of the ginger on the right?

Gold answer: yes

Fusion: left

✗ **incorrect**

Fusion (SigLIP): yes

✓ **correct**

Illustrative only. Representation quality is not isolated; the encoder and two widths changed together, so the example illustrates a system difference.



image 2401578

Question: What is the color of the letters?

Gold answer: blue

Fusion: yellow

✗ **incorrect**

Fusion (SigLIP): blue

✓ **correct**

Illustrative only. Representation quality is not isolated; the encoder and two widths changed together, so the example illustrates a system difference.

QC06_answer_side_disagreement

The pretrained 360M readout B4 and its random control B4r disagree. Development set, train_40k, seed 0. Condition: normal.

B4: the pretrained frozen SmoLM2-360M answer readout; B4r: the architecture-matched randomly initialised SmoLM2-360M answer readout, the within-size control for B4.



image 1159442

Question: Where is it?

Gold answer: street

B4 (pretrained): street

✓ correct

B4r (random): road

✗ incorrect

Illustrative only. The aggregate contrast is not directional and its interval contains zero. Both directions of disagreement must be shown, and the caption must not imply the pretrained side is better.



image 2368959

Question: Are there any people to the left of the man that is holding the camera?

Gold answer: yes

B4 (pretrained): no

✗ incorrect

B4r (random): yes

✓ correct

Illustrative only. The aggregate contrast is not directional and its interval contains zero. Both directions of disagreement must be shown, and the caption must not imply the pretrained side is better.

QC08_short_reasoning

A two-step-or-fewer question. Development set, train_40k, seed 0. Condition: normal.

fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings.



image 2396197

Question: Is the ground wet?

Gold answer: yes

GQA program steps: 2

Fusion: no

✗ **incorrect**

Illustrative only. Step count is the GQA semantic program length, a proxy for reasoning depth.



image 2347720

Question: Is the snowboard on the left of the picture?

Gold answer: yes

GQA program steps: 2

Fusion: yes

✓ **correct**

Illustrative only. Step count is the GQA semantic program length, a proxy for reasoning depth.